16. Java Stream Examples — Quick Notes

Sum of Array using Stream

```
int arr[] = {10,20,30};
int sum = Arrays.stream(arr).sum();
System.out.println(sum);
```

Output: 60

Important

Arrays.stream(arr)
creates stream over array.

Maximum Element using Stream

```
int arr[] = {10,20,30};
OptionalInt res =Arrays.stream(arr).max();
System.out.println(res.getAsInt());
```

Output: 30

Why OptionalInt?

Because:

- stream may be empty
- max may not exist

Methods:

- getAsInt()
- isPresent()

Average using Stream

```
int arr[] = {10,20,30};

double avg = Arrays.stream(arr)
        .average()
        .getAsDouble();

System.out.println(avg);
```

Output: 20.0

Infinite Stream Example

Problem: Print first 10 numbers containing digit 5.

```
Stream.iterate(1, x -> x + 1)
   .filter(x ->x.toString().contains("5"))
   .limit(10)
   .forEach(System.out::println);
```

Output: 5 15 25 35 45 50 51 52 53 54

Important Methods

Method	Purpose
iterate()	Infinite stream
filter()	Apply condition
limit()	Restrict output
forEach()	Print/process

String Filtering Example

Problem

Print strings:

- starting with a
- ending with c

```
String arr[] = {"abc","aec","xyz"};

Arrays.stream(arr)
  .filter(x -> x.startsWith("a"))
  .filter(x -> x.endsWith("c"))
  .sorted()
  .forEach(System.out::println);
```

Output:

```
abc
aec
```

Stream of Objects Example

Student Class

```
class Student {
  int roll;
  int marks;
  String name;
}
```

Average Marks Example

```
double avg = Arrays.stream(arr)
                .mapToInt(x -> x.getMarks())
                .average()
                .getAsDouble();
```

Output: 70.0

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mapToInt()

Convert: Stream<Student> **Into:** IntStream

Using: x -> x.getMarks()

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collect()

Collect stream data into:

- List
- Set
- Map

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toMap() Example

```
Map<Integer,String> m = Arrays.stream(arr)
                .collect(Collectors.toMap(Student::getRoll,Student::getName));
```

Result

Key	Value
110	abc
101	bcd

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groupingBy() Example: Group students by marks

```
Map<Integer,List<Student>> m =  Arrays.stream(arr)
                                .collect(Collectors.groupingBy(Student::getMarks));
```

Result Idea

Marks	Students
70	Student1, Student2
60	Student3

Common Stream Examples

Operation	Example
sum()	Total
average()	Average
max()	Maximum
filter()	Conditions
sorted()	Sorting
collect()	Store results
groupingBy()	Group objects

Important Quick Points

Topic	Point
Arrays.stream()	Stream over array
OptionalInt	Safe primitive optional
limit()	Restricts infinite stream
mapToInt()	Object stream → IntStream
collect()	Stream → Collection
toMap()	Create map
groupingBy()	Group objects
Stream Code	Concise & readable